

Chapter 12

Computer Vision

Computer vision encompasses computational methods for automatically understanding, processing, and analyzing visual data from the real world, ranging from reproducing human visual abilities to creating entirely new visual capabilities.

Human Vision vs Computer Vision:

- Human: Instantly recognizes faces, objects, scenes without conscious effort
- Computer: Needs complex algorithms to achieve basic visual understanding
- **Like** teaching a blind person to “see” through mathematical reasoning

Need

- Automate visual tasks that are effortless for humans but challenging for computers
- Process the massive amounts of visual data generated daily
- Enable applications from medical imaging to autonomous vehicles

Example Applications

- Object recognition: “This is a cat”
- Face detection: “Find all faces in this photo”
- Optical character recognition: “Read this text”
- Medical imaging: “Detect tumors in X-rays”

Preprocessing in Computer Vision

Preprocessing involves standardizing and transforming raw image data into a canonical form that reduces unwanted variability and makes learning easier for deep networks.

Key Operations

1. **Pixel Normalization:** Scale pixel values to consistent range [0,1] or [-1,1]
2. **Resizing:** Crop or scale images to standard dimensions
3. **Dataset Augmentation:** Artificially expand training data with transformations

Need

- Different images have varying scales, lighting, and formats
- Neural networks perform better with normalized, consistent input
- Reduces the burden on the model to learn invariances

Example

```
# Before preprocessing: pixels in [0, 255], various sizes  
# After preprocessing: pixels in [0, 1], fixed 224x224 size
```

Contrast Normalization

Contrast normalization standardizes the magnitude of differences between bright and dark pixels in an image, typically by setting the standard deviation of pixel intensities to a constant value.

Mathematics

Global Contrast Normalization (GCN):

For image $\mathbf{X} \in \mathbb{R}^{r \times c \times 3}$:

$$\mathbf{X}'_{i,j,k} = s \frac{\mathbf{X}_{i,j,k} - \bar{\mathbf{X}}}{\max \left\{ \epsilon, \sqrt{\lambda + \frac{1}{3rc} \sum_{i=1}^r \sum_{j=1}^c \sum_{k=1}^3 (\mathbf{X}_{i,j,k} - \bar{\mathbf{X}})^2} \right\}}$$

Where:

- $\bar{X}$ is the mean intensity
- s is the target standard deviation
- λ is a regularization parameter
- ϵ prevents division by zero

Types

1. **Global Contrast Normalization:** Normalizes entire image
2. **Local Contrast Normalization:** Normalizes within small windows

Need

- Images have varying lighting conditions and contrast levels
- Helps neural networks focus on shapes and structures rather than absolute intensities
- Places all images on similar scales

Example

Before GCN: Dark image vs bright image with different contrast ranges

After GCN: Both images have similar contrast characteristics

Dataset Augmentation

Dataset augmentation creates modified copies of existing training examples through class-preserving transformations to artificially increase dataset size and improve generalization.

Common Transformations

- Random translations and rotations
- Horizontal flipping
- Color perturbations
- Nonlinear geometric distortions

Need

- Limited training data leads to overfitting
- Teaches model to be invariant to irrelevant variations
- Effectively regularizes the model

Example

Original: Image of a cat

Augmented: Same cat rotated 15°, shifted left, slightly brighter

Speech Recognition

Automatic Speech Recognition (ASR) maps acoustic signals containing spoken language into corresponding word sequences by estimating:

$$f_{\text{ASR}}^*(\mathbf{X}) = \arg \max_y P^*(\mathbf{y} \mid \mathbf{X} = \mathbf{X})$$

Historical Evolution

1. **GMM-HMM Era (1980s-2009):** Gaussian Mixture Models + Hidden Markov Models
2. **Deep Learning Revolution (2009+):** Neural networks replacing GMMs
3. **End-to-End Systems:** Direct acoustic-to-word mapping

Need

- Enable human-computer interaction through speech
- Transcribe audio content automatically
- Power virtual assistants, transcription services, accessibility tools

Example

Input: Audio waveform “Hello world”

Output: Text “hello world”

Natural Language Processing (NLP)

NLP enables computers to use human languages through computational models that process, understand, and generate natural language text.

Core Tasks

- Machine translation
- Text classification
- Language modeling
- Sentiment analysis

Need

- Process the vast amount of textual data available
- Enable cross-lingual communication
- Automate text understanding and generation

Example

Input: English sentence

Output: French translation

n-grams Language Models

n -gram models define probability distributions over token sequences using fixed-length contexts, where each token’s probability depends only on the previous $n - 1$ tokens.

Mathematics

$$P(x_1, \dots, x_\tau) = P(x_1, \dots, x_{n-1}) \prod_{t=n}^{\tau} P(x_t \mid x_{t-n+1}, \dots, x_{t-1})$$

Types

- **Unigram** ($n = 1$): $P(x_t)$
- **Bigram** ($n = 2$): $P(x_t \mid x_{t-1})$
- **Trigram** ($n = 3$): $P(x_t \mid x_{t-1}, x_{t-2})$

Challenges

- **Curse of Dimensionality:** $|\mathbb{V}|^n$ possible n -grams
- **Sparsity:** Most n -grams never appear in training data
- **Smoothing Needed:** Handle zero probability events

Example

Trigram: $P(\text{away} \mid \text{dog ran}) = \frac{\text{count}(\text{"dog ran away"})}{\text{count}(\text{"dog ran"})}$

Neural Language Models (NLMs)

Neural language models overcome the curse of dimensionality using distributed word representations that enable sharing of statistical strength across similar words and contexts.

Key Innovation: Word Embeddings

- Map discrete words to continuous vector space
- Similar words have similar embeddings
- Enable generalization across semantically related contexts

Mathematics

Basic Architecture:

$$a_i = b_i + \sum_j W_{ij} h_j$$

$$\hat{y}_i = \frac{e^{a_i}}{\sum_{i=1}^{|\mathbb{V}|} e^{a_i}}$$

Need

- Overcome limitations of n -gram models
- Capture semantic relationships between words
- Handle large vocabularies efficiently

Example

Words “dog” and “cat” have similar embeddings because they appear in similar contexts.

High-Dimensional Outputs Challenge

The Problem

Softmax over large vocabularies requires $O(|\mathbb{V}|n_h)$ computations, which becomes prohibitive for $|\mathbb{V}| > 100,000$.

Solutions

1. Short List Approach

- Handle frequent words with neural net
- Handle rare words with n -gram model
- Combined probability: $P(y = i \mid C) = 1_{i \in \mathbb{L}} P(y = i \mid C, i \in \mathbb{L}) (1 - P(i \in \mathbb{T} \mid C)) + 1_{i \in \mathbb{T}} P(y = i \mid C, i \in \mathbb{T}) P(i \in \mathbb{T} \mid C)$

2. Hierarchical Softmax

- Organize words in a tree structure
- Reduce computation from $O(|\mathbb{V}|)$ to $O(\log |\mathbb{V}|)$
- Predict word probability via product of node probabilities

3. Importance Sampling

- Estimate gradients using negative samples
- Avoid computing all vocabulary scores
- Use proposal distribution q for efficient sampling

Comparative Analysis

Domain	Primary Challenge	Key Technique	Mathematical Foundation
Computer Vision	Visual variability and scaling	Contrast normalization & augmentation	$\mathbf{X}' = s \frac{\mathbf{X} - \bar{\mathbf{X}}}{\sigma + \epsilon}$
Speech Recognition	Acoustic-to-linguistic mapping	Deep sequence models	$f_{\text{ASR}}^*(\mathbf{X}) = \arg \max_{\mathbf{y}} P^*(\mathbf{y} \mid \mathbf{X})$
N-gram Models	Data sparsity	Fixed-context counting	$P(x_t \mid x_{t-n+1}, \dots, x_{t-1}) = \frac{P_n(x_{t-n+1}, \dots, x_t)}{P_{n-1}(x_{t-n+1}, \dots, x_{t-1})}$
Neural Language Models	Vocabulary scalability	Distributed representations	$\hat{y}_i = \frac{e^{b_i + \sum_j W_{ij} h_j}}{\sum_k e^{b_k + \sum_j W_{kj} h_j}}$

Key Innovations Summary

1. **Computer Vision:** From hand-crafted features to learned representations through contrast normalization and augmentation
2. **Speech Recognition:** Evolution from GMM-HMM to end-to-end deep learning models
3. **Language Modeling:** Transition from sparse n -gram counts to dense neural embeddings
4. **Scalability Solutions:** Hierarchical softmax and sampling methods for large vocabularies

All domains share the common theme of moving from hand-designed features and statistical methods to learned representations through deep neural networks.